EXPERIMENT 4

Academic Year: 2026-27 Year and Sem: TE V
Subject: DEV Lab Course Code: 2175119

PROBLEM STATEMENT:

To automate software provisioning and configuration management using Ansible on a single Ubuntu machine.

Requirements:

Ubuntu 22.04 LTS , Ansible , Internet Connection

THEORY:

What is Configuration Management?

Configuration Management is the process of managing servers automatically instead of configuring them manually.

Instead of logging into every server and installing software one by one, a configuration management tool performs all tasks automatically.

What is Ansible?

Ansible is an open-source automation tool used for

- Software Provisioning
- Configuration Management
- Application Deployment
- Server Automation

It is developed by Red Hat.

Features of Ansible

- Agentless
- Uses SSH
- Simple YAML Syntax

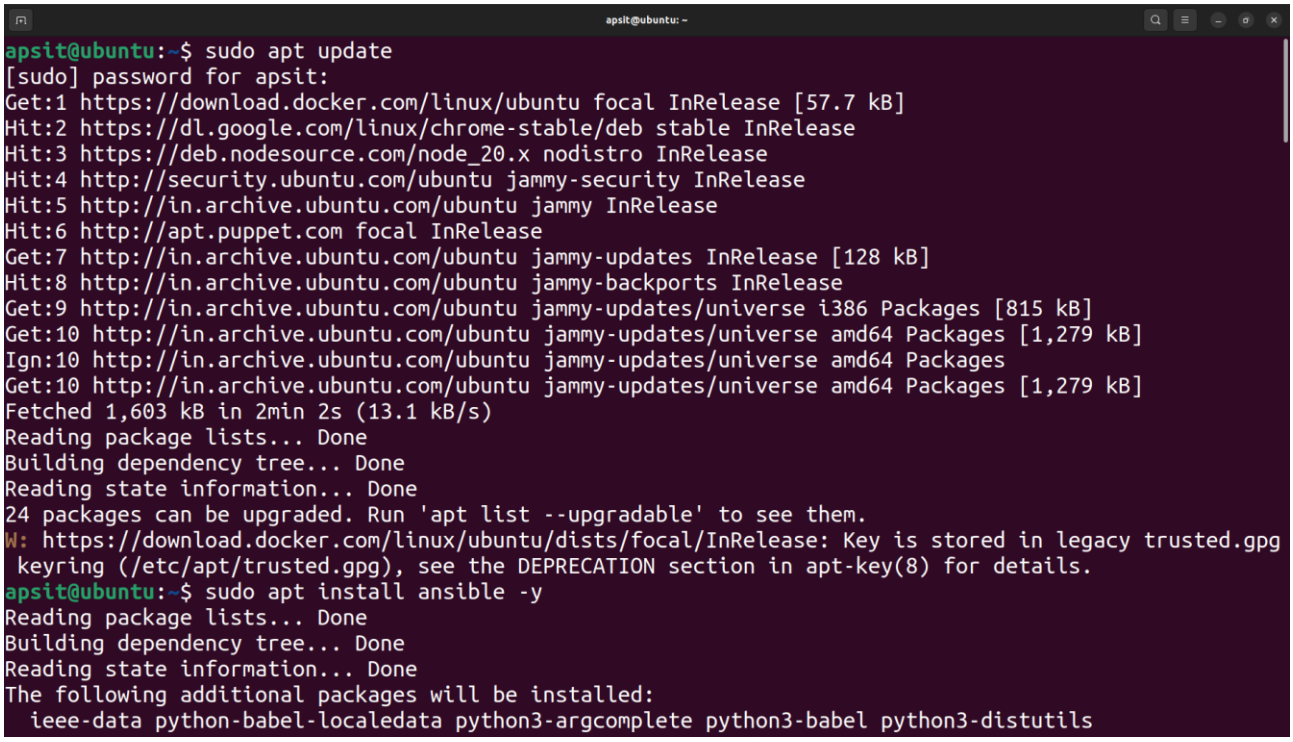
- Easy to Learn
- Idempotent
- Secure
- Cross Platform

STEPS FOR IMPLEMENTATION:

Step 1: Update Ubuntu

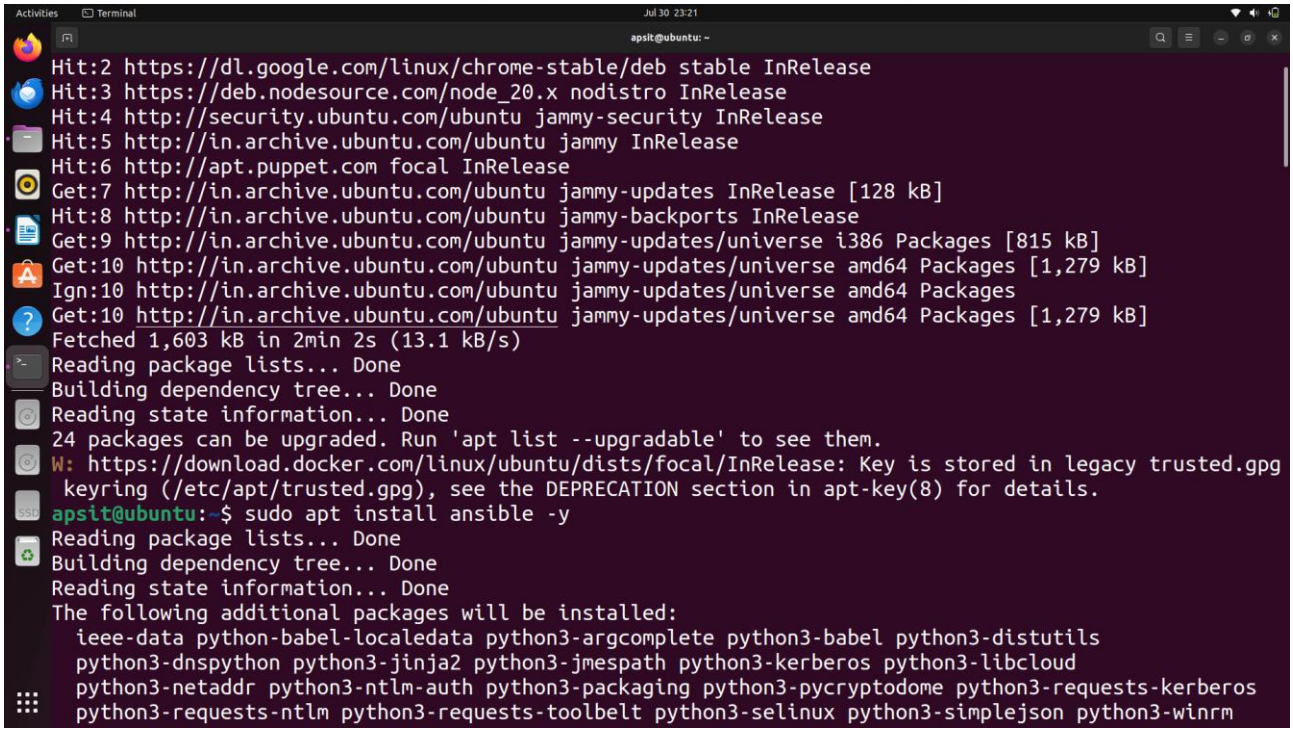
sudo apt update

sudo apt upgrade -y



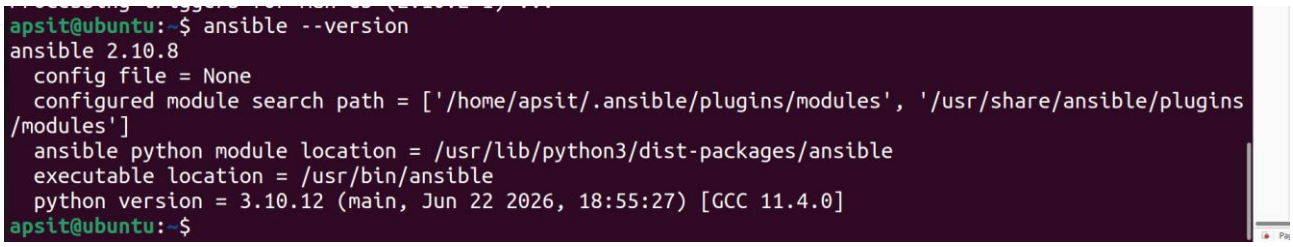
Step 2: Install Ansible

sudo apt install ansible -y



Step 3 :Verify installation

ansible --version



Step 4: Create Project Folder

mkdir ansible-lab

cd ansible-lab



Step 5: Create Inventory File

Create a file : nano inventory

Write :

[local]

localhost ansible_connection=local

Save the file.

Step 6: Test Ansible

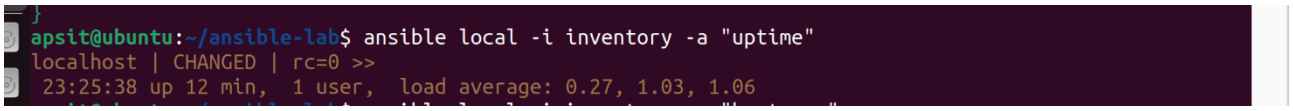
Run : ansible local -i inventory -m ping



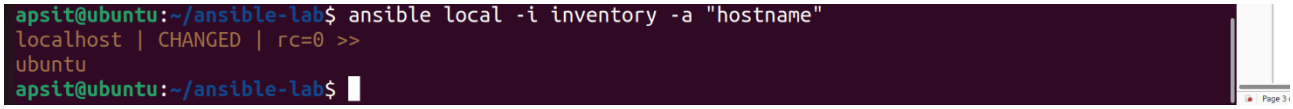
This confirms that Ansible is working correctly.

Step 7: Run Your First Command

Display system uptime. : ansible local -i inventory -a "uptime"



Display hostname.: ansible local -i inventory -a "hostname"



Display disk usage.: ansible local -i inventory -a "df -h"



Step 8: Create Your First Playbook

Create : nano install_apache.yml

```
Write

---

- name: Install Apache

  hosts: local

  become: yes

  tasks:

  - name: Install Apache

    apt:

      name: apache2

      state: present

      update_cache: yes
```

Save the file.

Step 9: Execute the Playbook

ansible-playbook -i inventory install_apache.yml

```
apsit@ubuntu:~/ansible-lab$ nano install_apache.yml
apsit@ubuntu:~/ansible-lab$ ansible-playbook -i inventory install_apache.yml

PLAY [Install Apache] ***********************************************************************

TASK [Gathering Facts] **********************************************************************
ok: [localhost]

TASK [Install Apache] ***********************************************************************

changed: [localhost]

PLAY RECAP **********************************************************************************
localhost                  : ok=2    changed=1    unreachable=0    failed=0    skipped=0    rescued=0
   ignored=0
```

Step 10: Verify Installation

systemctl status apache2

or

curl localhost

```
apsit@ubuntu:~/ansible-lab$ systemctl status apache2
● apache2.service - The Apache HTTP Server
     Loaded: loaded (/lib/systemd/system/apache2.service; enabled; vendor preset: enabled)
     Active: active (running) since Thu 2026-07-30 23:29:33 IST; 43s ago
       Docs: https://httpd.apache.org/docs/2.4/
   Main PID: 5897 (apache2)
      Tasks: 55 (limit: 9046)
     Memory: 5.6M
        CPU: 51ms
     CGroup: /system.slice/apache2.service
             ├─5897 /usr/sbin/apache2 -k start
             ├─5902 /usr/sbin/apache2 -k start
             └─5903 /usr/sbin/apache2 -k start

Jul 30 23:29:33 ubuntu systemd[1]: Starting The Apache HTTP Server...
Jul 30 23:29:33 ubuntu apachectl[5896]: AH00558: apache2: Could not reliably determine the server's f>
Jul 30 23:29:33 ubuntu systemd[1]: Started The Apache HTTP Server.
lines 1-16/16 (END)
```

Step 11: Create a Web Page Automatically

Create a playbook

nano webpage.yml

Write

- name: Create Web Page

```
hosts: local

become: yes

tasks:

- name: Copy HTML

  copy:

    dest: /var/www/html/index.html

    content: |

      <html>

      <h1>Welcome to Ansible Lab</h1>

      <h2>Automation Successful</h2>

      </html>
```

Run : ansible-playbook -i inventory webpage.yml

```
      ^ here
apsit@ubuntu:~/ansible-lab$ nano webpage.yml
apsit@ubuntu:~/ansible-lab$ nano webpage.yml
apsit@ubuntu:~/ansible-lab$ ansible-playbook -i inventory webpage.yml

PLAY [Create Web Page] *********************************************************************

TASK [Gathering Facts] *********************************************************************
ok: [localhost]

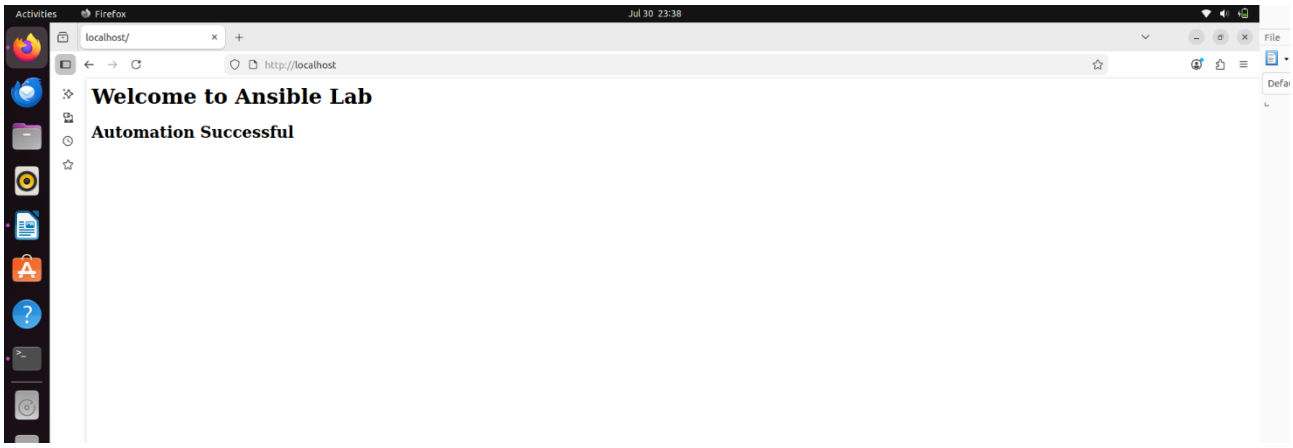
TASK [Copy HTML] ***************************************************************************
changed: [localhost]

PLAY RECAP *********************************************************************************
localhost                  : ok=2    changed=1    unreachable=0    failed=0    skipped=0    rescued=0
   ignored=0

apsit@ubuntu:~/ansible-lab$
```

Step 12: Open Browser

Visit : http://localhost



Step 13: Install Multiple Packages

Create : nano packages.yml

Write

```
---

- name: Install Packages


  hosts: local
```

```
become: yes

tasks:

- name: Install Multiple Packages

  apt:

    name:

      - git

      - tree

      - curl

    state: present
```

Run : ansible-playbook -i inventory packages.yml

```
apsit@ubuntu:~/ansible-lab$ nano packages.yml
apsit@ubuntu:~/ansible-lab$ ansible-playbook -i inventory packages.yml

PLAY [Install Packages] ********************************************************************

TASK [Gathering Facts] *********************************************************************
ok: [localhost]

TASK [Install Multiple Packages] ***********************************************************
changed: [localhost]

PLAY RECAP *********************************************************************************
localhost                  : ok=2    changed=1    unreachable=0    failed=0    skipped=0    rescued=0
   ignored=0

apsit@ubuntu:~/ansible-lab$
```

Verify

git --version

tree --version

curl –version

```
apsit@ubuntu:~/ansible-lab$ git --version
git version 2.34.1
apsit@ubuntu:~/ansible-lab$ tree --version
tree v2.0.2 (c) 1996 - 2022 by Steve Baker, Thomas Moore, Francesc Rocher, Florian Sesser, Kyosuke To
oro
apsit@ubuntu:~/ansible-lab$ curl --version
curl 7.81.0 (x86_64-pc-linux-gnu) libcurl/7.81.0 OpenSSL/3.0.2 zlib/1.2.11 brotli/1.0.9 zstd/1.4.8 li
idn2/2.3.2 libpsl/0.21.0 (+libidn2/2.3.2) libssh/0.9.6/openssl/zlib nghttp2/1.43.0 librtmp/2.3 OpenLD
P/2.5.20
Release-Date: 2022-01-05
Protocols: dict file ftp ftps gopher gophers http https imap imaps ldap ldaps mqtt pop3 pop3s rtmp rt
p scp sftp smb smbs smtp smtps telnet tftp
Features: alt-svc AsynchDNS brotli GSS-API HSTS HTTP2 HTTPS-proxy IDN IPv6 Kerberos Largefile libz NT
M NTLM_WB PSL SPNEGO SSL TLS-SRP UnixSockets zstd
apsit@ubuntu:~/ansible-lab$
```

Step 14: Manage Services

Create : nano apache_service.yml

Write

```
---

- name: Manage Apache

  hosts: local
```

become: yes

tasks:

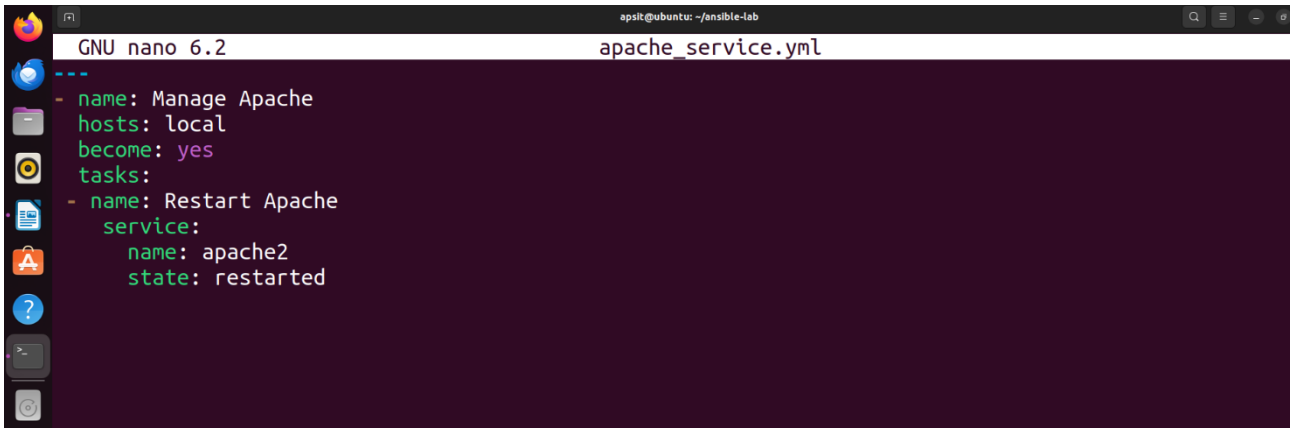
- name: Restart Apache

  service:

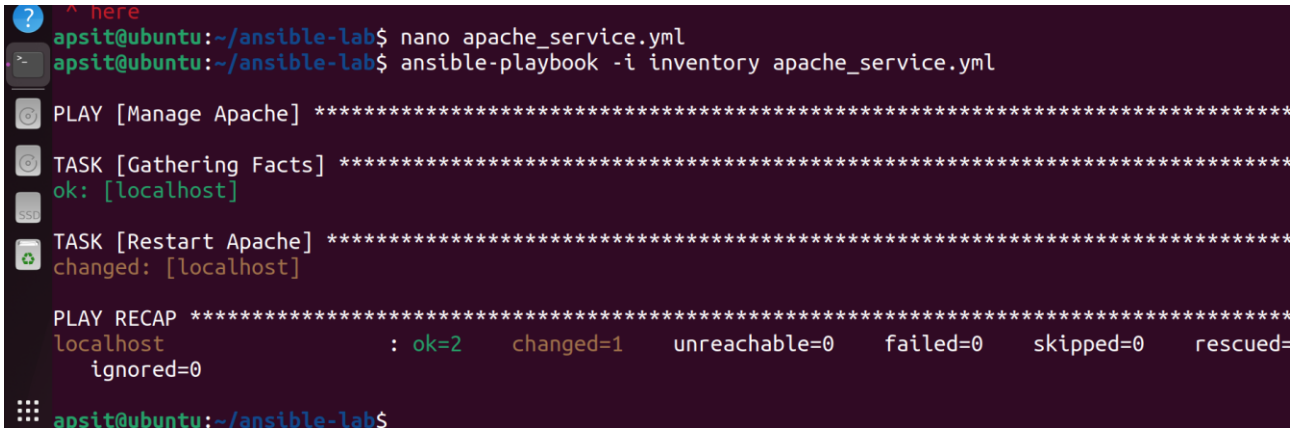
    name: apache2

    state: restarted

-

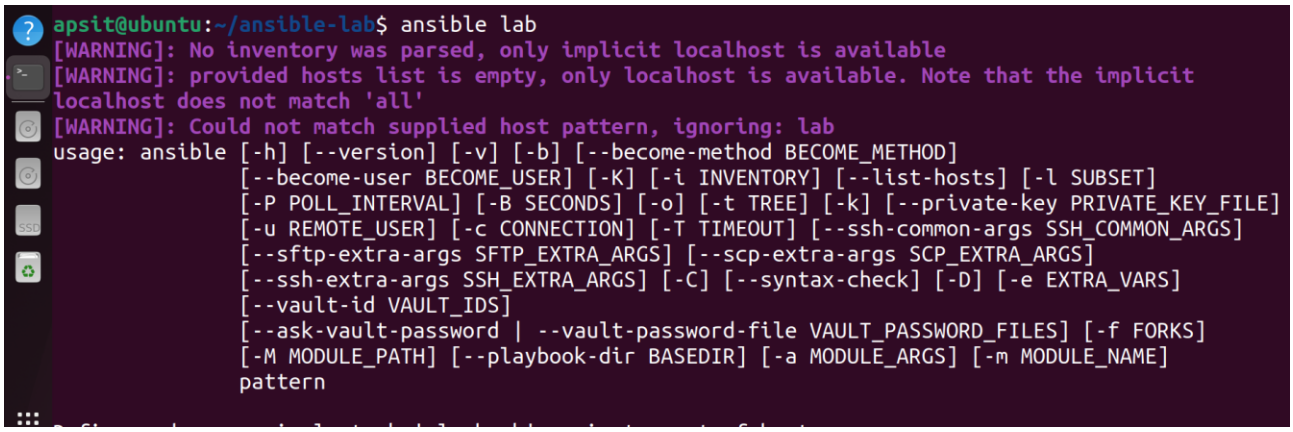


Run : ansible-playbook -i inventory apache_service.yml



Step 15 :Directory Structure

ansible-lab



│── inventory

│── install_apache.yml

│── webpage.yml

│── packages.yml

│── apache_service.yml

Conclusion : The experiment successfully demonstrated how to automate software provisioning and configuration management using Ansible on a single Ubuntu machine. By creating and executing an Ansible playbook, software installation and service management were performed automatically without manual intervention. This approach reduced configuration errors, improved consistency, and saved time by enabling repeatable and efficient system administration. The experiment also highlighted the agentless architecture of Ansible and its simplicity in managing system configurations, making it a powerful tool for DevOps automation.